SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO4	**Assessment:**	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	**Total Marks:**	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
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Date:	25-08-2026	**Duration:**	60 minutes

Code of Conduct

I, M Dhanush kumar (Reg. No. 192525318), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M Dhanush Kumar

Assigned Problem Statement: Smart Vertical Farming Automation Platform

The Smart Vertical Farming Automation Platform is designed to monitor hydrogen production, predict equipment failures, optimize energy usage, detect safety hazards, generate operational reports, and notify plant operators in real time. The platform is expected to increase production efficiency, reduce operational risks, lower maintenance costs, improve energy utilization, and enhance overall plant reliability. This document presents the test case design for the platform, covering requirement identification, test scenarios, detailed test cases, applicable test design techniques, and requirement traceability.

1. Requirements to be Tested

1.1 Functional Requirements

ID	Requirement
FR1	Collect and display real-time hydrogen production data from plant sensors.
FR2	Predict equipment failure using sensor trends and historical maintenance data.
FR3	Monitor energy consumption and recommend optimization actions.
FR4	Detect safety hazards (gas leakage, pressure, temperature) and raise alerts.
FR5	Generate periodic operational reports for the plant.
FR6	Notify operators of critical events through alerts/dashboard/SMS.
FR7	Authenticate operators and restrict access based on role.

1.2 Non-Functional Requirements

ID	Requirement
NFR1	Hazard alerts must be raised within 5 seconds of threshold breach.
NFR2	The platform must maintain at least 99.5% uptime.
NFR3	Sensor readings must be stored and displayed with no data loss.
NFR4	All operator access must be secured through authentication and role checks.
NFR5	The system must support concurrent monitoring of multiple sensor stations.

2. Test Scenarios

- Sensor data collection during normal plant operation.
- Sensor data collection when a sensor fails to report data.
- Production monitoring at the normal and critical threshold limits.
- Equipment failure prediction with sufficient historical sensor data.
- Equipment failure prediction when historical data is insufficient.
- Energy consumption monitoring and generation of optimization suggestions.
- Hazard detection when a single safety threshold is exceeded.
- Hazard detection when multiple hazards occur simultaneously.
- Delivery of operator notifications for critical events.
- Handling of notification delivery failure.

- Generation of scheduled and on-demand operational reports.
- Operator login with valid and invalid credentials.
- Access control enforcement for restricted operator actions.
- Platform behaviour when a monitoring service becomes unavailable.

3. Test Cases

The table below presents the detailed test cases derived from the scenarios in Section 2, covering normal, boundary, invalid, and exceptional conditions. Actual Result and Status columns are intentionally left for completion during execution.

TC ID	Requirement/Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC01	Production monitoring – valid sensor data	Start monitoring module; send reading from production sensor within normal range.	H2 output = 450 Nm3/hr	Reading is accepted and displayed on the dashboard in real time.	To be filled during execution	To be filled during execution
TC02	Production monitoring – missing sensor data	Disconnect a production sensor; observe monitoring module for 30 seconds.	No data received from Sensor-04	System flags sensor as 'No Data' and logs a data-loss event.	To be filled during execution	To be filled during execution
TC03	Production monitoring – invalid sensor value	Send a reading outside the physically possible range.	H2 output = -50 Nm3/hr	Reading is rejected and an invalid-data error is logged.	To be filled during execution	To be filled during execution
TC04	Production monitoring – value at normal limit	Send a reading exactly at the configured normal operating limit.	H2 output = 500 Nm3/hr (limit)	Reading is accepted; status remains 'Normal'.	To be filled during execution	To be filled during execution
TC05	Production monitoring – value at critical boundary	Send a reading exactly at the critical threshold.	H2 output = 600 Nm3/hr (critical limit)	System raises a 'Critical Production Level' alert.	To be filled during execution	To be filled during execution
TC06	Failure prediction – prediction generated	Feed 30 days of vibration/temperature trend data showing degradation.	Compressor bearing temp trending upward over 30 days	System generates a failure-prediction alert with recommended maintenance date.	To be filled during execution	To be filled during execution
TC07	Failure prediction – unavailable	Query prediction module with less than the minimum required history.	Only 2 days of sensor history	System returns 'Insufficient data for prediction' instead of a false result.	To be filled during execution	To be filled during execution
TC08	Energy monitoring – high consumption detected	Send energy readings above the configured efficiency threshold.	Energy use = 120% of baseline	System flags high energy consumption and logs the deviation.	To be filled during execution	To be filled during execution
TC09	Energy optimization – suggestion generated	Run optimization routine on a period with fluctuating load.	7-day energy and load dataset	System generates at least one actionable optimization recommendation.	To be filled during execution	To be filled during execution
TC10	Hazard detection – threshold exceeded	Send a gas-leak sensor reading above the safety threshold.	H2 concentration = 4.2% (threshold 4.0%)	System immediately raises a safety hazard alert.	To be filled during execution	To be filled during execution
TC11	Hazard detection – multiple hazards simultaneously	Send gas-leak and over-pressure readings at the same time.	Gas = 4.5%, Pressure = 105% of rated limit	System raises alerts for both hazards without suppressing either one.	To be filled during execution	To be filled during execution

TC ID	Requirement/Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC12	Notification – successfully sent	Trigger a critical hazard alert with notification service active.	Hazard alert event ID = HZ-1001	Operator receives notification on dashboard and registered channel within 5 seconds.	To be filled during execution	To be filled during execution
TC13	Notification – delivery failure	Trigger an alert while the notification/SMS gateway is offline.	Hazard alert event ID = HZ-1002, gateway down	System retries delivery and logs a notification-failure event for follow-up.	To be filled during execution	To be filled during execution
TC14	Reporting – operational report generation	Request a daily operational report for a completed shift.	Report period = 24-Aug-2026, Shift 1	Report is generated with production, energy, and hazard summaries.	To be filled during execution	To be filled during execution
TC15	Reporting – no data available for period	Request a report for a period with no recorded sensor data.	Report period = future date	System displays 'No data available' instead of an empty or broken report.	To be filled during execution	To be filled during execution
TC16	Access control – invalid login	Attempt operator login with an incorrect password.	Username = op_ravi, Password = wrong123	Login is rejected with an 'Invalid credentials' message; account is not locked on first attempt.	To be filled during execution	To be filled during execution
TC17	Access control – unauthorized access	Log in as a viewer-role operator and attempt to modify hazard thresholds.	Role = Viewer, Action = Edit Threshold	System denies the action and logs an unauthorized-access attempt.	To be filled during execution	To be filled during execution
TC18	System availability – service unavailable	Stop the sensor-gateway service while monitoring is active.	Sensor-gateway service = stopped	Dashboard displays a 'Service Unavailable' status instead of stale or false data.	To be filled during execution	To be filled during execution

4. Test Design Techniques

Not every technique was applied to every test case; each technique was used where it was most relevant to the type of input or behaviour being verified.

Technique	Justification / Applicability	Applied To
Equivalence Partitioning	Sensor and login input values were grouped into valid and invalid classes (e.g., valid production range vs. negative/out-of-range readings; correct vs. incorrect credentials) so that one representative value per class is tested instead of every possible value.	TC01, TC03, TC16
Boundary Value Analysis	Production levels and hazard thresholds are most likely to fail exactly at their limits, so values at and just beyond the normal and critical limits were specifically tested.	TC04, TC05, TC10
Decision Table	Hazard detection depends on combinations of conditions (gas level, pressure, multiple simultaneous readings); a decision-table style approach ensured each hazard combination produces the correct independent alert.	TC10, TC11
State Transition Testing	Equipment and prediction status move through defined states (Normal → Warning → Failure-Predicted) and operator sessions move through (Logged-out → Logged-in → Unauthorized), so transitions between these states were verified explicitly.	TC06, TC07, TC17

5. Traceability and Test Coverage

Requirement	Related Test Cases	Coverage
FR1 – Production monitoring	TC01–TC05	Full
FR2 – Failure prediction	TC06, TC07	Full
FR3 – Energy optimization	TC08, TC09	Full
FR4 – Hazard detection	TC10, TC11	Full
FR5 – Reporting	TC14, TC15	Full
FR6 – Operator notification	TC12, TC13	Full
FR7 – Access control	TC16, TC17	Full
NFR2 – Availability	TC18	Partial

All functional requirements (FR1–FR7) are fully covered by at least two test cases each, spanning normal, boundary, invalid, and exceptional conditions. The non-functional availability requirement (NFR2) is partially covered through the service-unavailable scenario; further coverage would require dedicated uptime/load testing beyond the scope of this document. Overall, the test case suite provides strong requirement coverage for the Smart Vertical Farming Automation Platform.

Conclusion

The designed test cases provide systematic coverage of the Smart Vertical Farming Automation Platform, including functional requirements, boundary conditions, invalid inputs, access control, notifications, reporting, and system availability. The test suite helps ensure that the platform operates reliably, securely, and efficiently for automated monitoring and management of vertical farming operations.